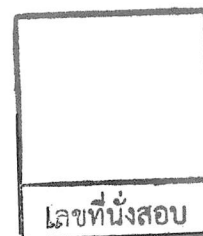


College of Industrial Technology
King Mongkut's University of Technology North Bangkok



Final Examination of Semester 1

Year: 2018

Subject: 353152 Electronic Circuits I

Section: 5-6

Date: 6 December 2018

Time: 13.00-15.00

Name: _____ ID: _____ Field of Study: _____

Instructions:

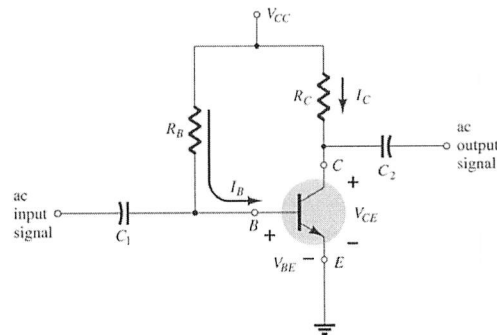
1. The examination has 11 pages (including this page), 2 sections (31 questions) and a total score of 80 points.
2. Write all your solutions and answers on this examination sheet.
3. This is a closed book examination.
4. You are not allowed to leave the exam room during the first 1 hour after the beginning of the exam.
5. You are not allowed to open the exam papers or start to answer before the proctor's permission.
6. You are not allowed to use the restroom during the exam except in case of an emergency.
7. No documents are allowed to be taken out of the examination room.
8. Calculator is allowed in the examination.
9. Electronic communication devices are NOT allowed in the examination room.
10. Cheating will result in failure of all classes registered for the current semester. Students who are caught cheating will also be denied registering for the following semester.

Cheating in the exam is considered an extremely serious offence which will result in expulsion from the University

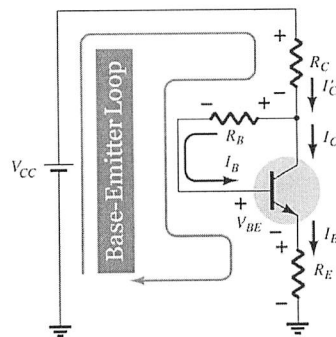
Part 1 (50 points) Multiple choices

Instruction: Mark the correct answer for the following questions in the provided answer sheet.

1. What is a bias type of the circuit in figure below.

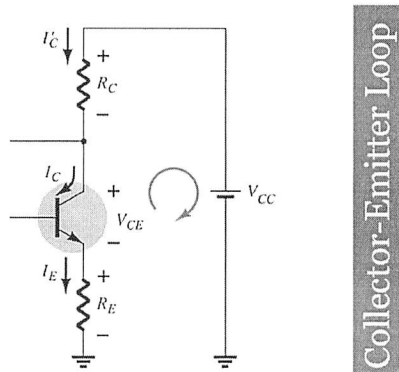


- (A) Fixed-bias circuit
(B) Emitter-stabilized bias circuit
(C) Voltage divider bias circuit
(D) DC bias with voltage feedback
2. Which of following transistor operating mode represents the short circuit?
- (A) cut-off (B) saturation (C) active (D) open
3. From Base-Emitter loop below, find the voltage equation of this closed loop.



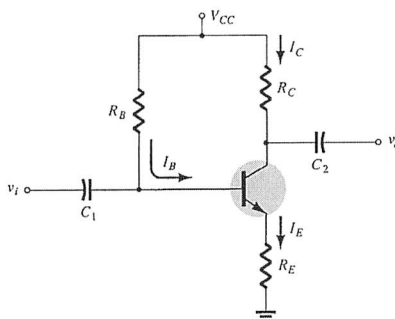
- (A) $I'_C R_C - I_B R_B - V_{BE} - I_E R_E = V_{CC}$
(B) $I'_C R_C - V_{BE} - I_E R_E = V_{CC}$
(C) $I'_C R_C + I_B R_B + V_{BE} + I_E R_E = V_{CC}$
(D) $I'_C R_C + V_{BE} + I_E R_E = V_{CC}$
4. In Collector-Feedback configuration, we assume that the $I'_C \cong I_C$, $I_E \cong I_C$. What is the reason of the assumption?
- (A) Because of $I_B \ll I_C$
(B) Because of $I_B = I_C$
(C) Because of $I_B \gg I_C$
(D) Because of $I_B \approx I_C$

5. From Collector-Emitter loop below, find the voltage equation of this closed loop.



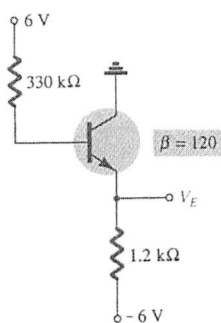
- (A) $I_E R_E + V_{CE} + I_E R_C = V_{CC}$ (B) $I_E R_E - V_{CE} - I_E R_C = V_{CC}$
 (C) $I_E R_E - V_{CE} + I_E R_C = V_{CC}$ (D) $I_E R_E - V_{CE} + I_E R_C = -V_{CC}$

6. The capacitor in circuit below are employed to.....



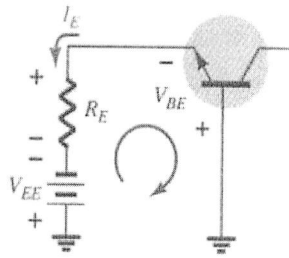
- (A) Mix DC and AC signal (B) Increase voltage level
 (C) Temporary source of voltage (D) Block DC

7. From the circuit below, What is the current direction of I_B, I_E ?



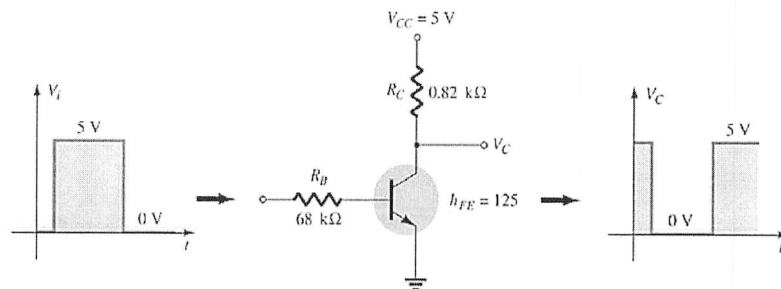
- (A) $I_B \downarrow, I_E \downarrow$ (B) $I_B \uparrow, I_E \uparrow$
 (C) $I_B \uparrow, I_E \downarrow$ (D) $I_B \downarrow, I_E \uparrow$

8. From the closed loop below, find the voltage equation of this closed loop.



- (A) $V_{EE} + I_E R_E = V_{BE}$ (B) $V_{EE} - I_E R_E = -V_{BE}$
 (C) $V_{EE} + I_E R_E = -V_{BE}$ (D) $V_{EE} - I_E R_E = V_{BE}$

9. From figure below, why output signal of this circuit is inverted.



- (A) 0V at input, the transistor is turn off and then Vout is Maximum.
 (B) 0V at input, the transistor is turn on and then Vout is Maximum.
 (C) 5V at input, the transistor is off and then Vout is Maximum.
 (D) 5V at input, the transistor is off and then Vout is Minimum.
10. The _____ stability factor, the _____ sensitive the network is to variations in that parameter.
 (A) higher, more (B) higher, less
 (C) lower, more (D) lower, less
11. Which class of power amplifiers that conducts less than 180° of input?
 (A) class A (B) class B (C) class AB (D) class C
12. Which class of power amplifiers having Q-point at the cutoff point?

- (A) class A (B) class B (C) class AB (D) class C
13. Which type of amplifier uses pulse (digital) signals in its operation?
 (A) Class A (B) Class B or AB (C) Class C (D) Class D
14. How can the class A amplifier efficiency be improved from 25%?
 (A) increase input current. (B) reduce input current.
 (C) add transformer at input side. (D) add transformer at output side.
15. What is the benefit of using Darlington pair and feedback pair transistors for push-pull operation?
 (A) reduce output swing (B) increase output power
 (C) increase input current (D) reduce distortion
16. What is the cause of crossover distortion in push-pull circuit?
 (A) two transistors do not turn on/off simultaneously.
 (B) heat at transistor.
 (C) two transistors are not the same type (PNP and NPN).
 (D) input current is the sine wave.
17. What is the value of I_{DC} of class B amplifier if peak output current $I_L(p)$ is 10 A.
 (A) 0.318 A (B) 0.636 A (C) 0.707 A (D) 1.414 A
18. What is the value of $P_o(AC)$ of class B amplifier if $V_L(p-p) = 10$ V and $R_L = 2$ ohms.
 (A) 50 W (B) 25 W (C) 6.25 W (D) 5.25 W
19. What is the purpose of using push-pull circuit for class B amplifier?
 (A) to obtain full cycle (B) to reduce the distortion
 (C) to increase speed of signal (D) to reduce the cost of amplifier
20. Which of the following is (are) the application(s) of a transistor?
 (A) Signal protector (B) Voltage regulator
 (C) Computer logic circuitry (D) DC Adapter

21. The emitter resistor in an emitter-stabilized bias circuit appears to be _____ in the base circuit.

- (A) larger (B) smaller
(C) the same (D) None of the above

22. Changes in temperature will affect the level of _____.

- (A) current gain β (B) leakage current I_{CEO}
(C) both current gain β and leakage current I_{CEO} (D) None of the above

23. In any amplifier employing a transistor, the collector current I_C is sensitive to _____.

- (A) β (B) V_{BE} (C) I_{CO} (D) V_{CE}

24. In a voltage-divider circuit, which one of the stability factors has the least effect on the device at very high temperature?

- (A) $S(I_{CO})$ (B) $S(V_{BE})$ (C) $S(\beta)$ (D) Undefined

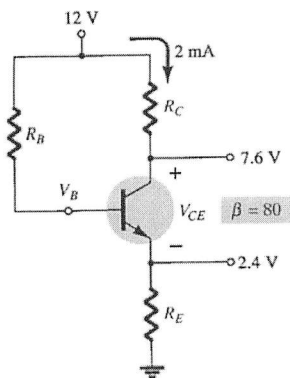
25. Which bias type circuit is the worst stability factor?

- (A) Fixed bias (B) Emitter-stabilized bias
(C) Voltage divider (D) Voltage feedback

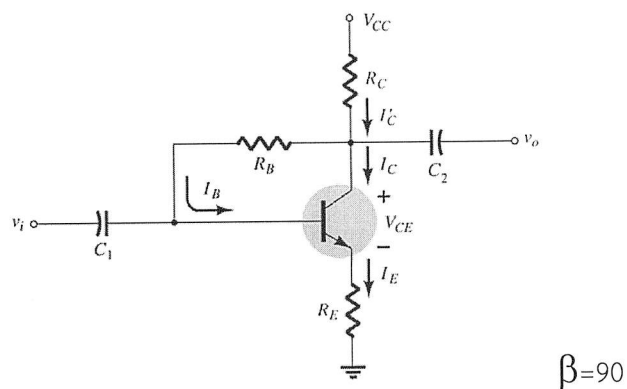
Part 2 (30 points) Calculation

Instruction: Show the mathematical expression and answers of following problems.

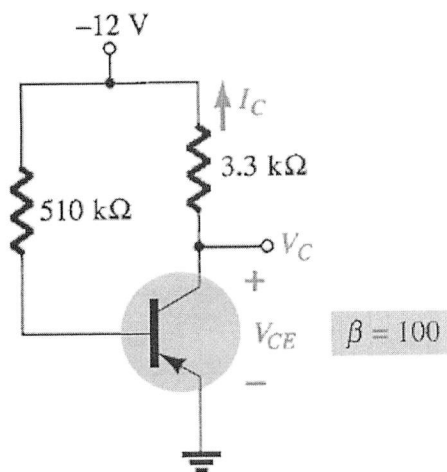
1. (5 points) From figure below, determine R_E .



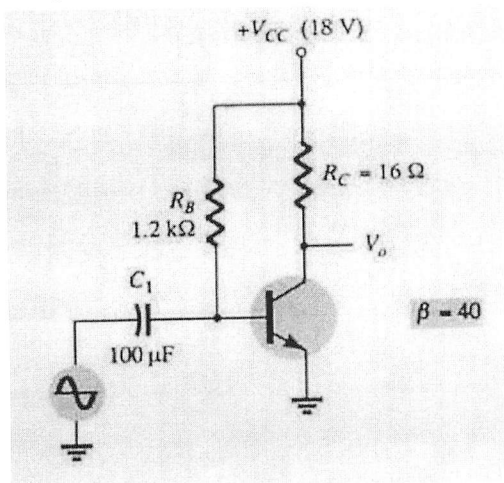
2. (5 points) Given $R_C=5k\Omega$, $R_B=200k\Omega$, $R_E=1.5k\Omega$ and $V_{CC}=12V$, find the current I_B and voltage V_{CE} .



3. (5 points) Determine the V_C , V_{CE} and I_C



4. (5 points) Determine the input and output power of the circuit below. Given the current of input signal is 5 mA_{rms} .



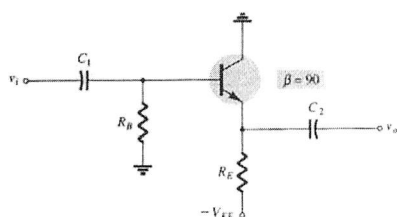
5. (5 points) For a class B amplifier with $V_{CC}=25\text{ V}$ driving an $8\ \Omega$ load, determine:

- 5.1 Maximum input power
- 5.2 Maximum output power
- 5.3 Maximum circuit efficiency

6. (5 points) For transformer-coupled class A amplifier, calculate the transformer turns ratio “a” required to connect a $4\ \Omega$ speaker so that they appear as an $8\ k\Omega$ effective load.

Handout

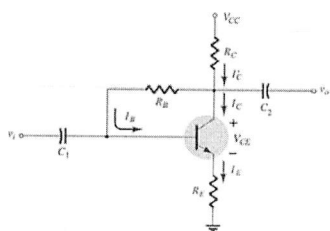
Emitter-follower



$$V_{CE} = V_{EE} - I_E R_E$$

$$I_B = \frac{V_{EE} - V_{BE}}{R_B + (\beta + 1)R_E}$$

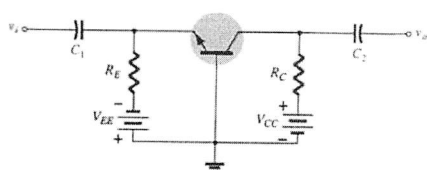
Collector-feedback



$$I_B = \frac{V_{CC} - V_{BE}}{R_B + \beta(R_C + R_E)}$$

$$V_{CE} = V_{CC} - I_C(R_C + R_E)$$

Common-base



$$I_E = \frac{V_{EE} - V_{BE}}{R_E}$$

$$V_{CE} = V_{EE} + V_{CC} - I_E(R_C + R_E)$$

Class A amplifier

$$P_i(DC) = \frac{V_{CC}^2}{2R_C}$$

$$P_o(AC) = \frac{V_{CC}^2}{8R_C}$$

Transformer-couple class A amplifier

$$\% \eta = 50 \left(\frac{V_{CE(max)} - V_{CE(min)}}{V_{CE(max)} + V_{CE(min)}} \right)^2$$

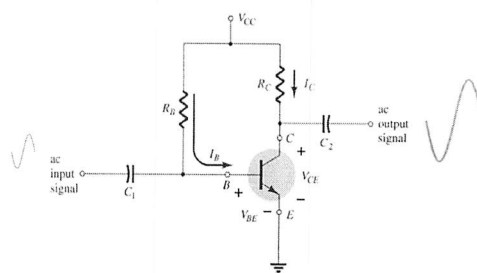
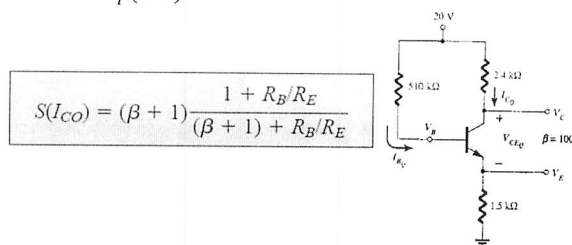
$$\frac{R'_L}{R_L} = \frac{R_1}{R_2} = \left(\frac{N_1}{N_2} \right)^2 = a^2$$

Class B amplifier

$$P_o(AC) = \frac{V_L^2(p-p)}{8R_L} = \frac{V_L^2(p)}{2R_L}$$

$$P_i(DC) = V_{CC} I_{dc} = V_{CC} \left(\frac{2}{\pi} I_L(p) \right)$$

$$\% \eta = \frac{P_o(ac)}{P_i(dc)} \times 100\%$$



$$I_{C_Q} = \beta I_{B_Q} \quad I_{B_Q} = \frac{V_{CC} - V_{BE}}{R_B}$$

Answer Sheet

Name: _____ ID: _____

Subject: 342152 Electronic Circuits I



	(A)	(B)	(C)	(D)
1.				
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